

Chapter 3: Diagnostic Human Learning Dimensions

The pipeline extracts six orthogonal diagnostic learning dimensions sequentially per student timeline to quantify latent psychometric behavior.

1 Learning Status (True Velocity State $v_{i,t}$)

- **Theoretical Foundation:** Dynamic Linear Modeling (DLM) & Kalman Filtering Theory.
- **Mathematical Execution:** Fit a discrete-time Local Linear Trend (LLT) state-space model onto the student's longitudinal `Actual_Leveli,t` curve. The state equation splits capability into level $\theta_{i,t}$ and velocity $v_{i,t}$:

$$\begin{bmatrix} \theta_{i,t} \\ v_{i,t} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \theta_{i,t-1} \\ v_{i,t-1} \end{bmatrix} + \begin{bmatrix} w_{1,t} \\ w_{2,t} \end{bmatrix}, \quad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{W}) \quad (1)$$

- **Formula:**

$$\text{Learning_Status}_{i,t} = v_{i,t} \quad (2)$$

- **Pedagogical Meaning:** Represents true instantaneous skill velocity (pages/class), stripped of observation noise.

2 Stability (Prediction Error Volatility $\sigma_{w,i,t}$)

- **Theoretical Foundation:** Stochastic Volatility Modeling & Bayesian Predictive Error Analysis.
- **Mathematical Execution:** Compute the moving standard deviation of one-step-ahead prediction residuals $\epsilon_{i,t} = \text{Actual_Level}_{i,t} - \hat{\mathbb{E}}[\text{Actual_Level}_{i,t} \mid \mathbf{X}_{i,t-1}]$ over a rolling 3-week window.

- **Formula:**

$$\text{Stability}_{i,t} = \sigma_{w,i,t} = \sqrt{\frac{1}{3} \sum_{\tau=t-2}^t (\epsilon_{i,\tau} - \bar{\epsilon}_{i,t})^2} \quad (3)$$

- **Pedagogical Meaning:** Quantifies lesson-to-lesson preparation consistency. Low volatility indicates disciplined home preparation; high volatility signals erratic practice habits.

3 Talent (Gaussian Normalized Dynamic Capacity Talent_{*i,t*})

- **Theoretical Foundation:** Latent Trait Psychometrics & Parametric Growth Curve Modeling.
- **Mathematical Execution:** Model talent as a dynamic parameter discovered chronologically over time. At step t , compute running intrinsic capacity $\Lambda_{i,t}$ based on cumulative defect resistivity:

$$\Lambda_{i,t} = \gamma \cdot \left(1.0 - \frac{1}{t} \sum_{\tau=1}^t \text{Defect_Density}_{i,\tau} \right) \quad (4)$$

where $\text{Defect_Density}_{i,\tau} = \frac{1}{18} \sum_{p=1}^{18} D_{i,\tau,p}$ and $\gamma = 0.2776$.

- **Population Z-Score Normalization:** To eliminate ceiling saturation and match downstream regression scales, normalize $\Lambda_{i,t}$ across active cohort observations at step t :

$$\text{Talent}_{i,t} = \frac{\Lambda_{i,t} - \mu_{\Lambda,t}}{\sigma_{\Lambda,t}} \quad (5)$$

- **Pedagogical Meaning:** Tracks evolving technical defect resistance relative to peers, stabilizing onto a personal capacity anchor.

4 Teacher Influence (Causal Instruction Absorption Elasticity β_{teacher})

- **Theoretical Foundation:** Instrumental Variable Causal Inference & Cross-Correlation Elasticity.
- **Mathematical Execution:** Compute the rolling regression slope (β) tracking co-movement between teacher instruction injections at $t - 1$ (`special_points_lagged`) and defect corrections at t :

$$\beta_{\text{teacher},i,t} = \frac{\text{Cov}(\text{Defect_Density}_{i,t}, \text{special_points}_{i,t-1})}{\text{Var}(\text{special_points}_{i,t-1}) + 1 \times 10^{-6}} \quad (6)$$

using a rolling 5-week window, normalized globally into Z-scores.

- **Pedagogical Meaning:** Measures how effectively a student absorbs and implements teacher corrections during home practice.

5 Effort (Latent Residual Drive $\Omega_{i,t}$)

- **Theoretical Foundation:** Structural Equation Modeling & Econometric Production Frontier Residuals.
- **Mathematical Execution:** Fit a cohort-wide expectation model mapping expected level change to Talent and teacher push:

$$\hat{\Delta}\text{Actual_Level}_{i,t} = \beta_1 \cdot \text{Talent}_{i,t} + \beta_2 \cdot (\text{Teacher_Influence}_{i,t} \cdot \text{special_points}_{i,t-1}) \quad (7)$$

- **Formula:**

$$\text{Effort_Raw}_{i,t} = \Delta\text{Actual_Level}_{i,t} - \hat{\Delta}\text{Actual_Level}_{i,t} \quad (8)$$

normalized globally into standard Z-scores ($\Omega_{i,t}$).

- **Pedagogical Meaning:** Captures unobserved personal exertion and willpower exceeding baseline capacity expectations.

6 Efficiency (Quality-Adjusted Output ROI $\Theta_{i,t}$)

- **Theoretical Foundation:** Microeconomic Production Frontier Modeling & Resource Allocation Theory.
- **Mathematical Execution:** Model efficiency as output capability return per unit of exertion weighted by error saturation:

- **Formula:**

$$\text{Efficiency_Raw}_{i,t} = \frac{\max(0, \Delta\text{Actual_Level}_{i,t})}{\max(0.1, \text{Effort_Raw}_{i,t}) \cdot (1.0 + \text{Defect_Density}_{i,t})} \quad (9)$$

normalized globally into standard Z-scores ($\Theta_{i,t}$).

- **Pedagogical Meaning:** Quantifies practice quality. High efficiency indicates smart, focused practice; low efficiency indicates mindless, error-prone grinding.